Signed sub-optimality bound for sample-path AIS

Extended Theorem 3

Lemma 1 (Policy evaluation)**.** *Let $\hat{\pi} = (\hat{\pi}_1, \cdots, \hat{\pi}_T)$, $\hat{\pi}_t : \mathcal{Z} \to \mathcal{A}$, be any AIS policy and $\pi_t = \pi_t \circ \sigma$, $\pi_t : \mathcal{H} \to \mathcal{A}$ its lifted version with $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$ for all $t \in \{1, \cdots, T\}$. Define*

$$\alpha_t^{\hat{\pi}}(h_t) = V_t^{\pi}(h_t) - \hat{V}^{\hat{\pi}}(\sigma_t(h_t)), \tag{1}$$

$$\beta_t^{\hat{\pi}}(h_t, a_t) = Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t), \tag{2}$$

and

$$\Delta_t^{\hat{\pi}}(h_t, a_t) = \Big[c_t(h_t, a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1}|h_t, a_t)\Big] \tag{3}$$

$$- \Big[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1}|h_t, a_t)\Big]. \tag{4}$$

Then, for all $h_t \in \mathcal{H}_t$ and $a_t \in \mathcal{A}$, the following holds

$$\beta_t^{\hat{\pi}}(h_t, a_t) = \Delta_t^{\hat{\pi}}(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \tag{5}$$

$$\alpha_t^{\hat{\pi}}(h_t) = \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))). \tag{6}$$

where $\alpha_{T+1}^{\hat{\pi}} \equiv 0$.

Proof. We first establish the recursion (5) and then the identity (6).

Step 1 (the $\beta^{\hat{\pi}}$-recursion). Fix $t \in \{1, \cdots, T\}$, $h_t \in \mathcal{H}_t$, and $a_t \in \mathcal{A}$. Adding and subtracting $\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} \mid h_t, a_t)$, and using $\hat{P}_t(\,\cdot \mid \sigma_t(h_t), a_t) = \hat{P}_t^{\sigma}(\,\cdot \mid h_t, a_t)$, we get

$$\begin{aligned}
\beta_t^{\hat{\pi}}(h_t, a_t) &= Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t) \\
&= \big[c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)\big] \\
&\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t(dz_{t+1} \mid \sigma_t(h_t), a_t) \\
&= \underbrace{\big[c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)\big] + \textstyle\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1}|h_t, a_t) - \textstyle\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1}|h_t, a_t)}_{\Delta_t^{\hat{\pi}}(h_t, a_t)} \\
&\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1}|h_t, a_t),
\end{aligned}$$

where the first bracket is $\Delta_t^{\hat{\pi}}(h_t, a_t)$ by (3). For the remaining terms, we have

$$\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} \mid h_t, a_t) = \int \hat{V}_{t+1}^{\hat{\pi}}\big(\sigma_{t+1}(h_{t+1})\big) P_{Y,t}(dy_{t+1} \mid h_t, a_t).$$

Thus the last term can be written as

$$\int V^{\pi}_{t+1}(h_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t)$$
$$-\int \hat{V}^{\hat{\pi}}_{t+1}(z_{t+1})P^{\sigma}_t(dz_{t+1}|h_t,a_t) = \int \Big[V^{\pi}_{t+1}(h_{t+1}) - \hat{V}^{\hat{\pi}}_{t+1}\big(\sigma_{t+1}(h_{t+1})\big)\Big] P_{Y,t}(dy_{t+1} \mid h_t, a_t)$$
$$= \int \alpha^{\hat{\pi}}_{t+1}(h_t, a_t, y_{t+1})\, P_{Y,t}(dy_{t+1} \mid h_t, a_t)$$

so the two integrals against $P_{Y,t}$ combine into

$$\int \Big[V^{\pi}_{t+1}(h_{t+1}) - \hat{V}^{\hat{\pi}}_{t+1}\big(\sigma_{t+1}(h_{t+1})\big)\Big] P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \int \alpha^{\hat{\pi}}_{t+1}(h_t, a_t, y_{t+1})\, P_{Y,t}(dy_{t+1} \mid h_t, a_t),$$

This establishes (5). At $t = T$, we have $\alpha^{\hat{\pi}}_{T+1} \equiv 0$ and $\Delta^{\hat{\pi}}_T(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$, so (5) reads $\beta^{\hat{\pi}}_T = \Delta^{\hat{\pi}}_T$.

Step 2 (the $\alpha^{\hat{\pi}}$-identity). Since $\hat{\pi}_t$ is deterministic and $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$, the value functions are the Q-functions evaluated at the prescribed actions:

$$V^{\pi}_t(h_t) = Q^{\pi}_t\big(h_t, \hat{\pi}_t(\sigma_t(h_t))\big), \qquad \hat{V}^{\hat{\pi}}_t(\sigma_t(h_t)) = \hat{Q}^{\hat{\pi}}_t\big(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))\big).$$

Then, we can write

$$\alpha^{\hat{\pi}^\star}_t(h_t) = V^{\pi}_t(h_t) - \hat{V}^{\hat{\pi}}(\sigma_t(h_t)) \tag{7}$$
$$= Q^{\pi}_t\big(h_t, \hat{\pi}_t(\sigma_t(h_t))\big) - \hat{Q}^{\hat{\pi}}_t\big(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))\big) \tag{8}$$
$$= \beta^{\hat{\pi}}_t\big(h_t, \hat{\pi}_t(\sigma_t(h_t))\big) \tag{9}$$

□

Lemma 2 (Optimality)**.** *Define*

$$\alpha^\star_t(h_t) = V^\star_t(h_t) - \hat{V}^\star_t(\sigma_t(h_t)), \tag{10}$$
$$\beta^\star_t(h_t, a_t) = Q^\star_t(h_t, a_t) - \hat{Q}^\star_t(\sigma_t(h_t), a_t), \tag{11}$$

and

$$\Delta_t(h_t, a_t) = \Big[c_t(h_t, a_t) + \int \hat{V}^\star_{t+1}(z_{t+1})P^{\sigma}_t(dz_{t+1}|h_t, a_t)\Big] \tag{12}$$
$$- \Big[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}^\star_{t+1}(z_{t+1})\hat{P}^{\sigma}_t(dz_{t+1}|h_t, a_t)\Big]. \tag{13}$$

Then, the following holds

$$\beta^\star_t(h_t, a_t) = \Delta_t(h_t, a_t) + \int \alpha^\star_{t+1}(h_t, a_t, y_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t), \tag{14}$$
$$\beta^\star_t(h_t, \pi^\star_t(h_t)) \le \alpha^\star_t(h_t) \le \beta^\star_t(h_t, \hat{\pi}^\star_t(\sigma_t(h_t))). \tag{15}$$

where $\alpha^\star_{T+1} \equiv 0$.

Proof. Step 1 (the $\beta^\star$-recursion). [Complete Step 1, similar to Lemma 1.1]

The derivation is identical to Step 1 in the proof of Lemma 1 with $(Q_t^\pi, \hat{Q}_t^{\hat{\pi}}, V_{t+1}^\pi, \hat{V}_{t+1}^{\hat{\pi}})$ replaced by $(Q_t^\star, \hat{Q}_t^\star, V_{t+1}^\star, \hat{V}_{t+1}^\star)$, add and subtract $\int \hat{V}_{t+1}^\star(z_{t+1}) P_t^\sigma(dz_{t+1} \mid h_t, a_t)$, and identify the cost-and-kernel-mismatch bracket as $\Delta_t(h_t, a_t)$ from (12).

Step 2 (the $\alpha^\star$-inequalities). The two inequalities in (15) follow from the two optimality properties, used one at a time.

Upper bound. By optimality of $\hat{\pi}_t^\star$ on the AIS side, $\hat{V}_t^\star(\sigma_t(h_t)) = \hat{Q}_t^\star\big(\sigma_t(h_t), \hat{\pi}_t^\star(\sigma_t(h_t))\big)$, while on the true side $V_t^\star(h_t) = \inf_{a \in \mathcal{A}} Q_t^\star(h_t, a) \leq Q_t^\star\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big)$. Therefore,

$$\begin{aligned}\alpha_t^\star(h_t) &= V_t^\star(h_t) - \hat{V}_t^\star(\sigma_t(h_t)) \\ &\leq Q_t^\star\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big) - \hat{Q}_t^\star\big(\sigma_t(h_t), \hat{\pi}_t^\star(\sigma_t(h_t))\big) = \beta_t^\star\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big).\end{aligned}$$

Lower bound. By optimality of $\pi_t^\star$ on the true side, $V_t^\star(h_t) = Q_t^\star\big(h_t, \pi_t^\star(h_t)\big)$, while on the AIS side $\hat{V}_t^\star(\sigma_t(h_t)) = \inf_{a \in \mathcal{A}} \hat{Q}_t^\star(\sigma_t(h_t), a) \leq \hat{Q}_t^\star\big(\sigma_t(h_t), \pi_t^\star(h_t)\big)$. Therefore,

$$\begin{aligned}\alpha_t^\star(h_t) &= V_t^\star(h_t) - \hat{V}_t^\star(\sigma_t(h_t)) \\ &\geq Q_t^\star\big(h_t, \pi_t^\star(h_t)\big) - \hat{Q}_t^\star\big(\sigma_t(h_t), \pi_t^\star(h_t)\big) = \beta_t^\star\big(h_t, \pi_t^\star(h_t)\big).\end{aligned}$$

Combining the two gives (15). □

Theorem 3. *Suppose that for each $t \in \{1, \cdots, T\}$, there exist measurable functions $\alpha_t^{\hat{\pi}^\star} : \mathcal{H}_t \to \mathbb{R}$, $\alpha_t^\star : \mathcal{H}_t \to \mathbb{R}$, $\beta_t^{\hat{\pi}^\star} : \mathcal{H}_t \times \mathcal{A} \to \mathbb{R}$, and $\beta_t^\star : \mathcal{H}_t \times \mathcal{A} \to \mathbb{R}$ such that*

$$\beta_T^{\hat{\pi}^\star}(h_T, a_T) \geq \Delta_T(h_T, a_T) \geq \beta_T^\star(h_T, a_T) \tag{16}$$

and for $t < T$:

$$\beta_t^{\hat{\pi}^\star}(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^\star}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \tag{17}$$

$$\beta_t^\star(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^\star(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \tag{18}$$

and for all $t \in \{1, \cdots, T\}$:

$$\alpha_t^{\hat{\pi}^\star}(h_t) \geq \beta_t^{\hat{\pi}^\star}\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big), \tag{19}$$

$$\alpha_t^\star(h_t) \leq \beta_t^\star\big(h_t, \pi_t^\star(h_t)\big), \tag{20}$$

where Δ_t is as defined in (12) (in particular, at terminal time, $\Delta_T(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$), and $\pi_t^\star$ and $\hat{\pi}_t^\star$ are as in Lemma 2. Then, for all $t \in \{1, \cdots, T\}$ and all $h_t \in \mathcal{H}_t$, the AIS-based policy $\pi_t(h_t) = \hat{\pi}_t^\star(\sigma_t(h_t))$ satisfies

$$V_t^\pi(h_t) - V_t^\star(h_t) \leq \alpha_t^{\hat{\pi}^\star}(h_t) - \alpha_t^\star(h_t). \tag{21}$$

Write the lower bound as well

Proof. [Check proof]

Throughout the proof, let $\tilde{\alpha}_t^{\hat{\pi}^\star}$, $\tilde{\beta}_t^{\hat{\pi}^\star}$, $\tilde{\alpha}_t^\star$, and $\tilde{\beta}_t^\star$ denote the *exact* signed error functions of Lemmas 1 and 2:

$$\tilde{\alpha}_t^{\hat{\pi}^\star}(h_t) = V_t^\pi(h_t) - \hat{V}_t^\star(\sigma_t(h_t)), \qquad \tilde{\beta}_t^{\hat{\pi}^\star}(h_t, a_t) = Q_t^\pi(h_t, a_t) - \hat{Q}_t^\star(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 1 specialized to $\hat{\pi} = \hat{\pi}^\star$, and

$$\tilde{\alpha}_t^\star(h_t) = V_t^\star(h_t) - \hat{V}_t^\star(\sigma_t(h_t)), \qquad \tilde{\beta}_t^\star(h_t, a_t) = Q_t^\star(h_t, a_t) - \hat{Q}_t^\star(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 2. Regarding the specialization $\hat{\pi} = \hat{\pi}^\star$ in Lemma 1: since $\hat{\pi}_t^\star(z_t)$ is a minimizer of $\hat{Q}_t^\star(z_t, \cdot)$, evaluating the AIS dynamic program along $\hat{\pi}^\star$ shows that the policy-evaluation value functions of $\hat{\pi}^\star$ coincide with the optimal ones, $\hat{V}_t^{\hat{\pi}^\star} = \hat{V}_t^\star$ and $\hat{Q}_t^{\hat{\pi}^\star} = \hat{Q}_t^\star$ for all t; consequently, the integrand $\hat{V}_{t+1}^{\hat{\pi}^\star}$ in (**??**) equals $\hat{V}_{t+1}^\star$ and the mismatch terms coincide, $\Delta_t^{\hat{\pi}^\star} = \Delta_t$, the same Δ_t as in Lemma 2 and in the statement of the theorem. Thus Lemmas 1 and 2 give, for all t, h_t, and a_t (with $\tilde{\alpha}_{T+1}^{\hat{\pi}^\star} \equiv 0$ and $\tilde{\alpha}_{T+1}^\star \equiv 0$):

$$\begin{aligned}
\tilde{\beta}_t^{\hat{\pi}^\star}(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^\star}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\
\tilde{\alpha}_t^{\hat{\pi}^\star}(h_t) &= \tilde{\beta}_t^{\hat{\pi}^\star}\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big), \qquad (22) \\
\tilde{\beta}_t^\star(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^\star(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\
\tilde{\beta}_t^\star\big(h_t, \pi_t^\star(h_t)\big) &\le \tilde{\alpha}_t^\star(h_t). \qquad (23)
\end{aligned}$$

Step 1 (the candidate functions compared to the exact errors). We show by backward induction that for all $t \in \{1, \cdots, T\}$ and all $h_t \in \mathcal{H}_t$,

$$\alpha_t^{\hat{\pi}^\star}(h_t) \ge \tilde{\alpha}_t^{\hat{\pi}^\star}(h_t) \qquad \text{and} \qquad \alpha_t^\star(h_t) \le \tilde{\alpha}_t^\star(h_t). \qquad (24)$$

Base case ($t = T$). Since $\tilde{\alpha}_{T+1}^{\hat{\pi}^\star} \equiv 0$ and $\tilde{\alpha}_{T+1}^\star \equiv 0$, the recursions in (22) and (23) give $\tilde{\beta}_T^{\hat{\pi}^\star} = \tilde{\beta}_T^\star = \Delta_T$. Hence, by (16), $\beta_T^{\hat{\pi}^\star} \ge \tilde{\beta}_T^{\hat{\pi}^\star}$ and $\beta_T^\star \le \tilde{\beta}_T^\star$. Therefore, by (19) and the identity in (22),

$$\alpha_T^{\hat{\pi}^\star}(h_T) \ge \beta_T^{\hat{\pi}^\star}\big(h_T, \hat{\pi}_T^\star(\sigma_T(h_T))\big) \ge \tilde{\beta}_T^{\hat{\pi}^\star}\big(h_T, \hat{\pi}_T^\star(\sigma_T(h_T))\big) = \tilde{\alpha}_T^{\hat{\pi}^\star}(h_T),$$

and, by (20) and the sandwich inequality in (23),

$$\alpha_T^\star(h_T) \le \beta_T^\star\big(h_T, \pi_T^\star(h_T)\big) \le \tilde{\beta}_T^\star\big(h_T, \pi_T^\star(h_T)\big) \le \tilde{\alpha}_T^\star(h_T).$$

Induction step ($t < T$). Suppose (24) holds at $t+1$. Then, by (17), the induction hypothesis, and the recursion in (22),

$$\begin{aligned}
\beta_t^{\hat{\pi}^\star}(h_t, a_t) &\ge \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^\star}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\
&\ge \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^\star}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^\star}(h_t, a_t),
\end{aligned}$$

and then, exactly as in the base case, $\alpha_t^{\hat{\pi}^\star}(h_t) \ge \beta_t^{\hat{\pi}^\star}\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big) \ge \tilde{\beta}_t^{\hat{\pi}^\star}\big(h_t, \hat{\pi}_t^\star(\sigma_t(h_t))\big) = \tilde{\alpha}_t^{\hat{\pi}^\star}(h_t)$. Symmetrically, by (18), the induction hypothesis, and the recursion in (23),

$$\begin{aligned}
\beta_t^\star(h_t, a_t) &\le \Delta_t(h_t, a_t) + \int \alpha_{t+1}^\star(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\
&\le \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^\star(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^\star(h_t, a_t),
\end{aligned}$$

and then $\alpha_t^\star(h_t) \le \beta_t^\star\big(h_t, \pi_t^\star(h_t)\big) \le \tilde{\beta}_t^\star\big(h_t, \pi_t^\star(h_t)\big) \le \tilde{\alpha}_t^\star(h_t)$. This establishes (24) for all t.

Step 2 (bridge identity). Adding and subtracting $\hat{V}_t^\star(\sigma_t(h_t))$,

$$V_t^\pi(h_t) - V_t^\star(h_t) = \big[V_t^\pi(h_t) - \hat{V}_t^\star(\sigma_t(h_t))\big] - \big[V_t^\star(h_t) - \hat{V}_t^\star(\sigma_t(h_t))\big] = \tilde{\alpha}_t^{\hat{\pi}^\star}(h_t) - \tilde{\alpha}_t^\star(h_t).$$

Combining Step 2 with (24),

$$V_t^{\pi}(h_t) - V_t^{\star}(h_t) = \tilde{\alpha}_t^{\hat{\pi}^\star}(h_t) - \tilde{\alpha}_t^{\star}(h_t) \leq \alpha_t^{\hat{\pi}^\star}(h_t) - \alpha_t^{\star}(h_t),$$

which is (21). $\square$